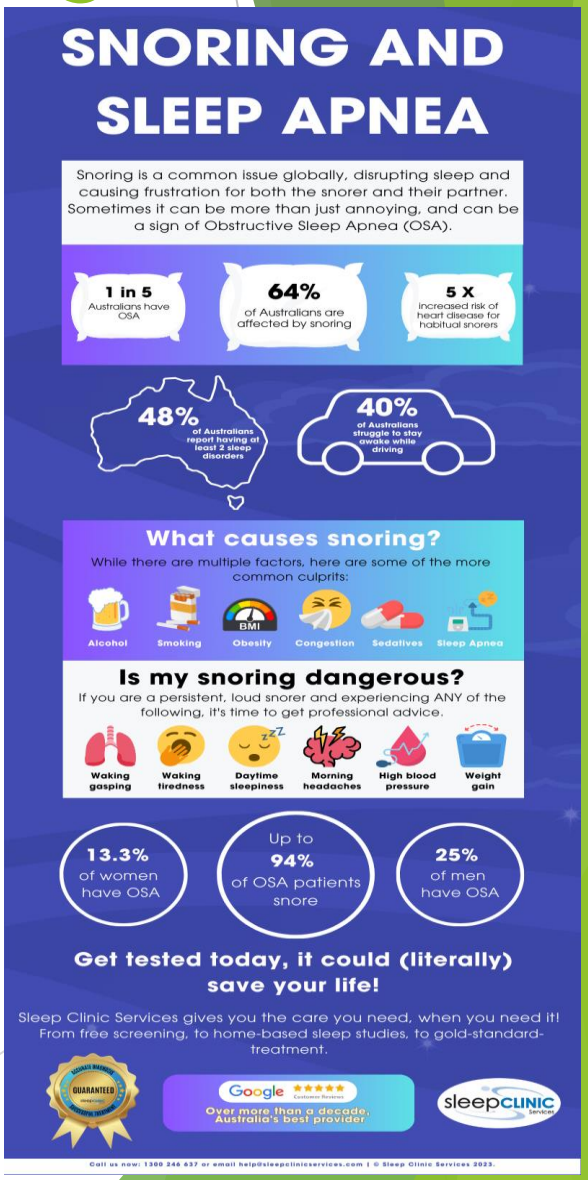
Noise-Robust Snore Detection Using Audio and Deep Learning

Introduction

- Snore detection: identifying snoring sounds
- Why it matters: sleep health, apnea detection

Clinical Relevance of Snoring

- Snoring prevalence: 20–30% adults, ~1B affected by OSA
- Linked to hypertension, heart disease, diabetes, cognitive issues
- Highlights medical + social importance



Problem Statement

- Goal: Classify audio clips as Snore vs Non-Snore
- Robustness against background noise

Research Gap & Contributions

- Gap: Most prior models failed in noisy conditions
- Contributions:
  - Noise-robust training
  - Model comparisons (Classical vs CNN vs YAMNet)
  - Edge deployment on Android
  - Explainability with SHAP

Data Collection

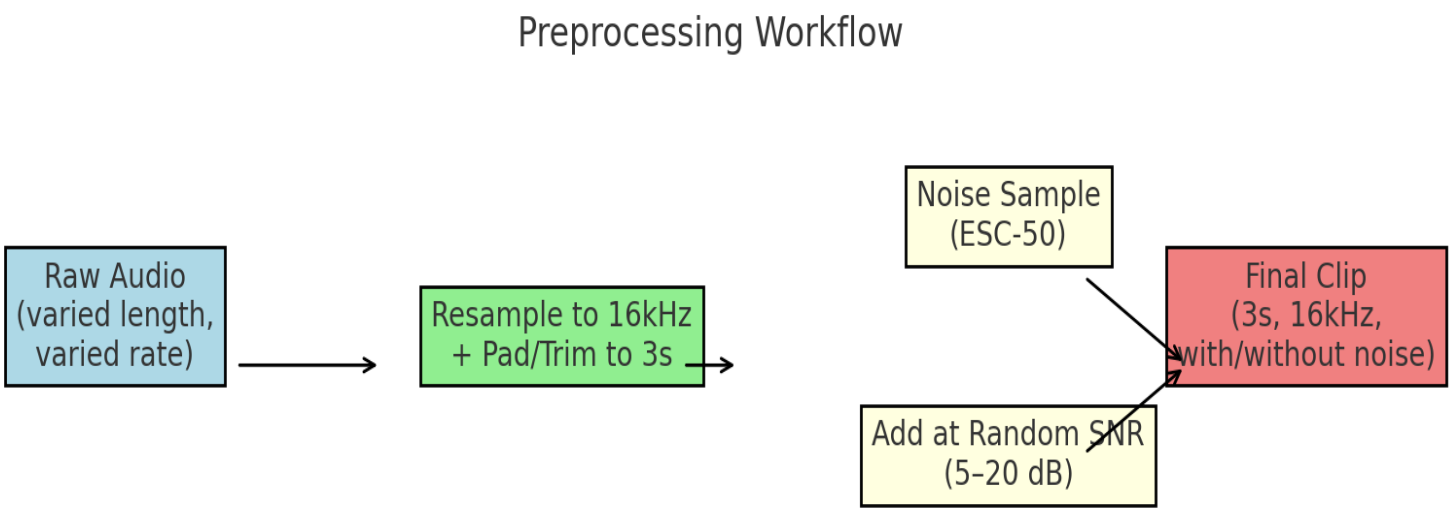
- Snore dataset (Kaggle): ~1000 clips (1s each)
  - Snore (500) and Non-snore (500)
- ESC-50 noises for augmentation

Classical Approaches

- MFCC, wavelet, entropy features used in older work
- Logistic Regression, SVM, Random Forest achieved moderate accuracy
- Limitation: Poor noise robustness, lab-only validation

Preprocessing

- Resample to 16kHz mono, pad/trim to 3s
- Add noise at random SNR

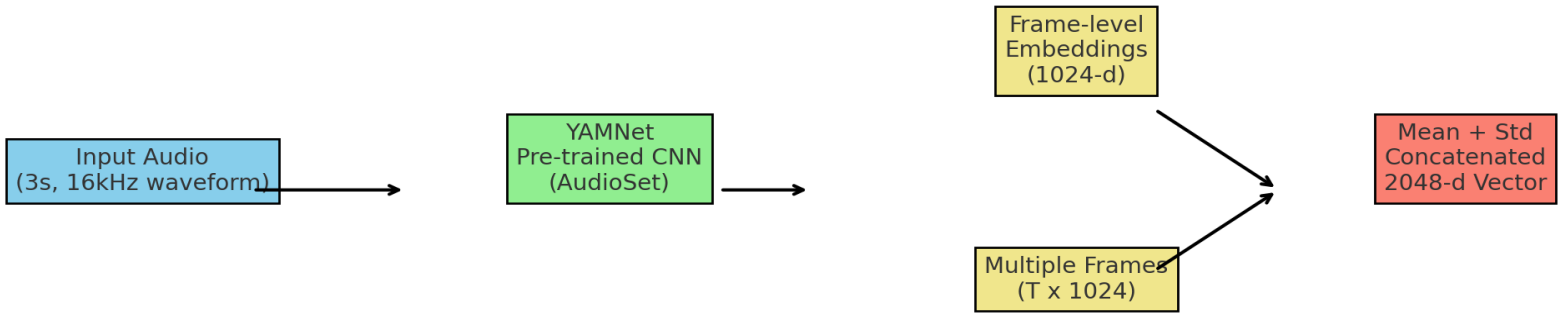


Deep Learning Advances

- CNNs, hybrid CNNs, transfer learning
- High accuracy on clean data (>90%)
- Weakness: Not systematically tested with noise

Feature Extraction

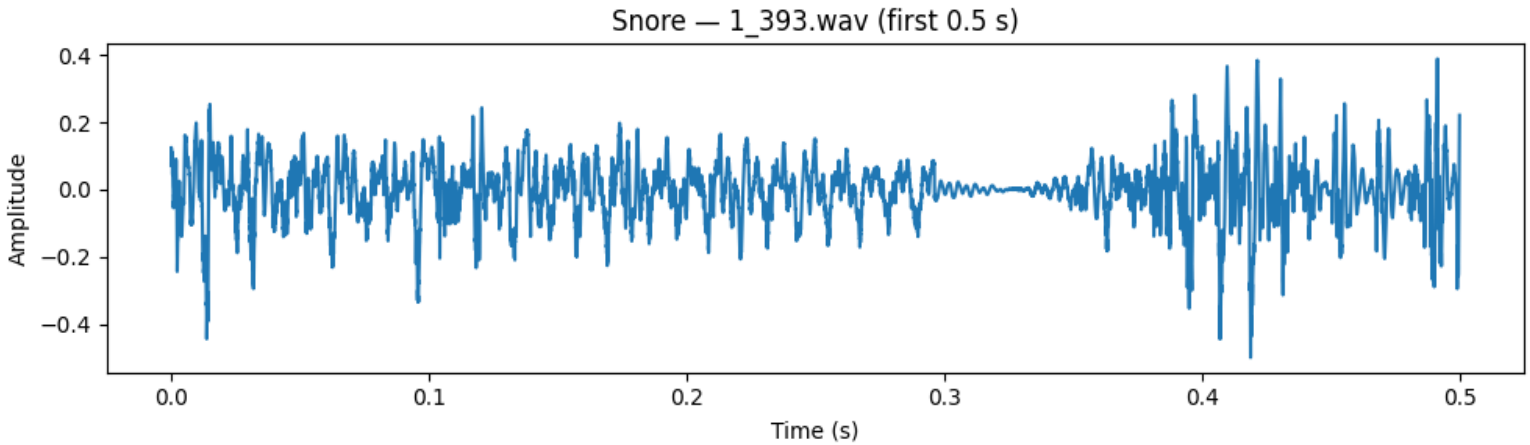
- Use YAMNet (pre-trained on Audio Set)
- Output 1024-dim embeddings per frame
- Take mean + std = 2048-dim vector



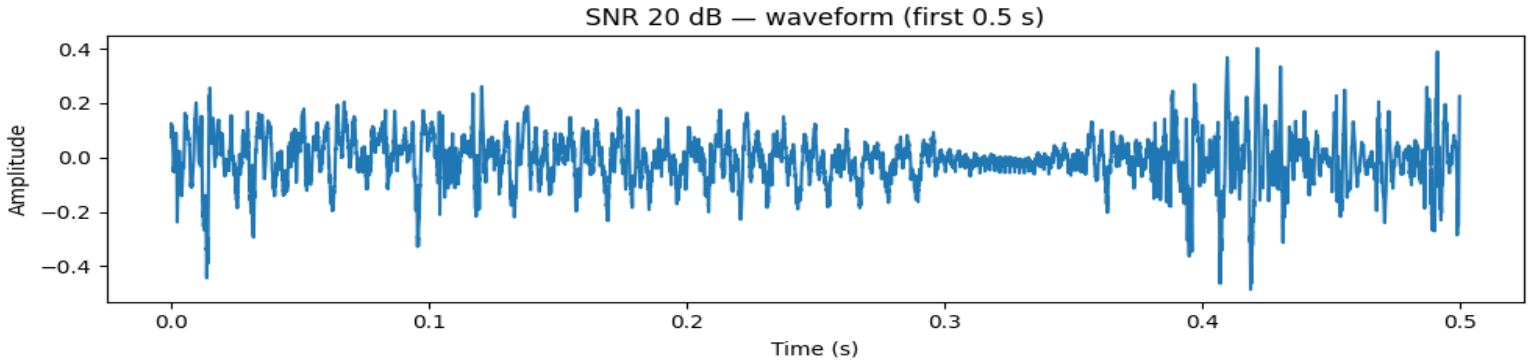
Data Augmentation

- Every training clip mixed with noise
- SNR varied -10 - 14 dB

Clean



Noisy

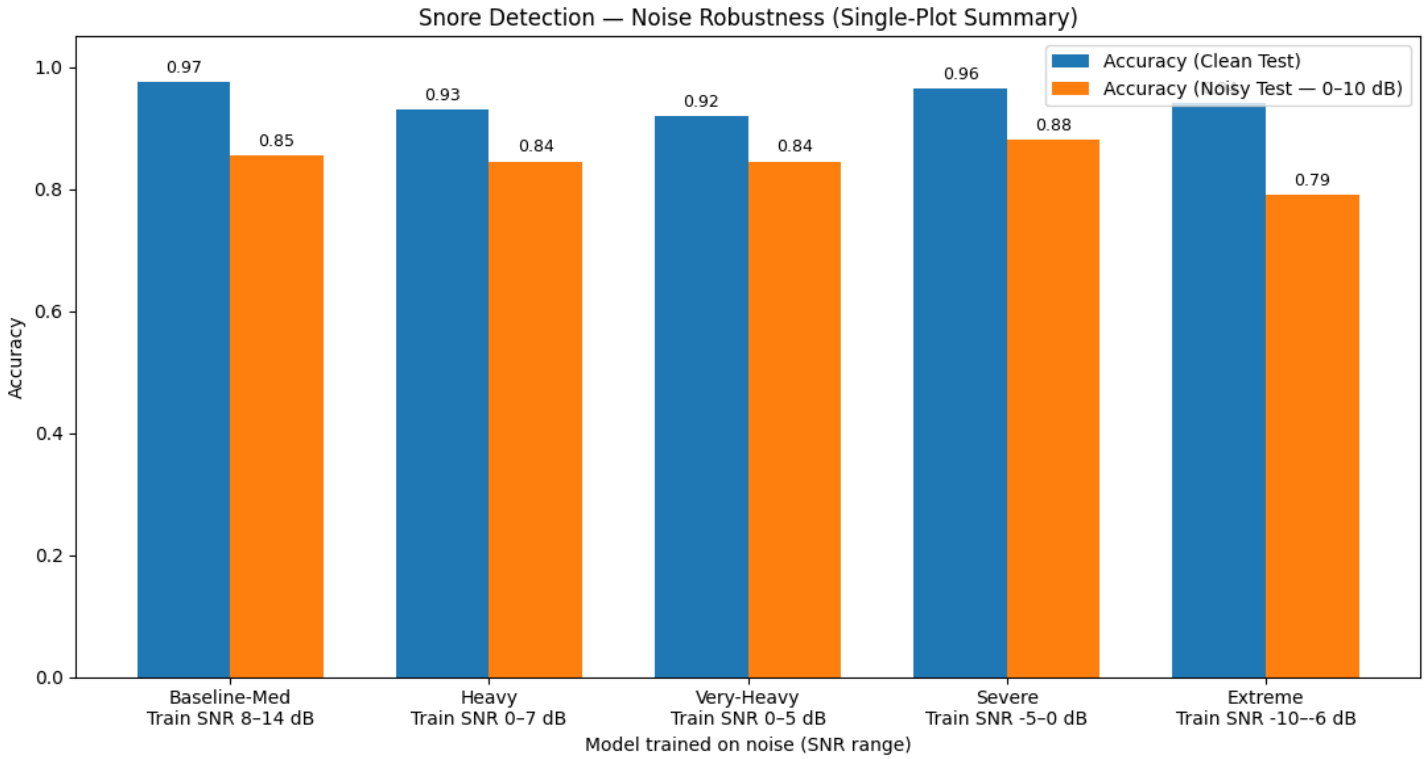


Noise Augmentation Strategy (Continued)

- Five regimes: Moderate (8–14dB), Heavy (0–7dB), Very-Heavy (0–5dB), Severe (–5–0dB), Extreme (–10 to –6dB)
- Key finding: Severe regime balanced clean and noisy performance (88% accuracy)

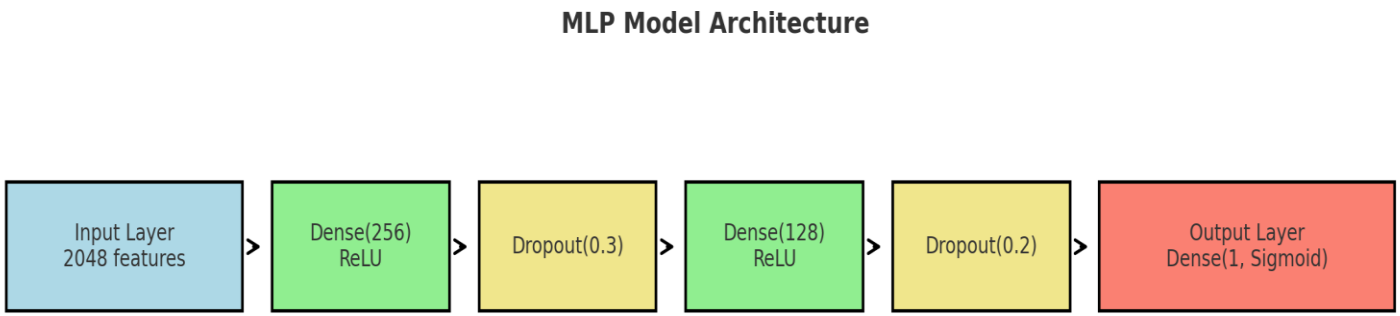
Noise Augmentation Strategy

- Comparison



Model Architecture

- MLP: Input 2048 → Dense(256, ReLU) → Dropout(0.3) → Dense(128, ReLU) → Dropout(0.2) → Dense(1, Sigmoid)

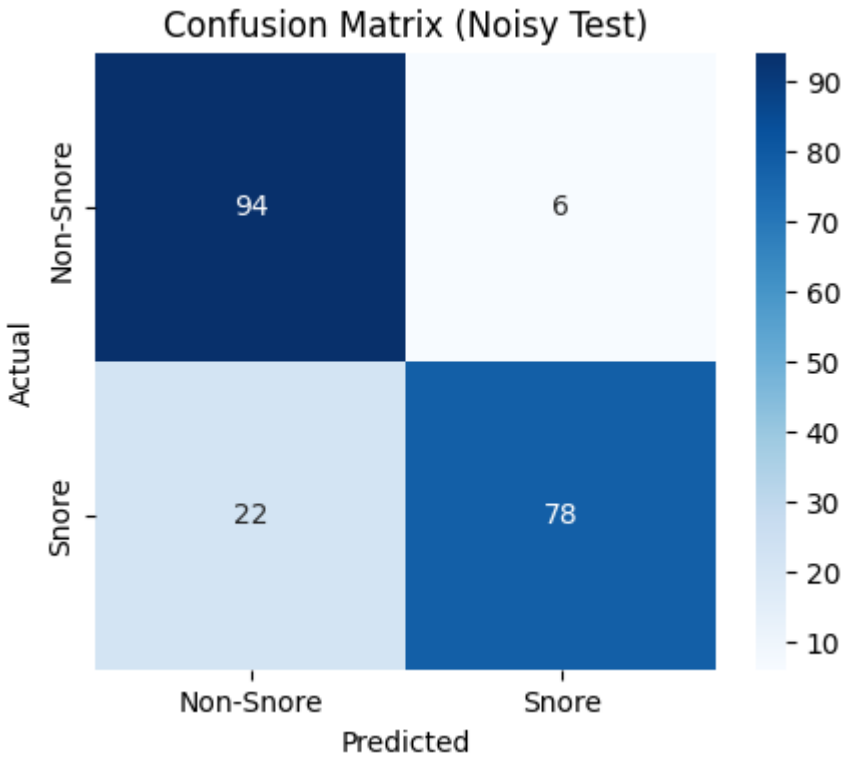


Training Strategy

- 80/20 train/test split
- 15% validation split
- Adam optimizer, early stopping, LR schedule

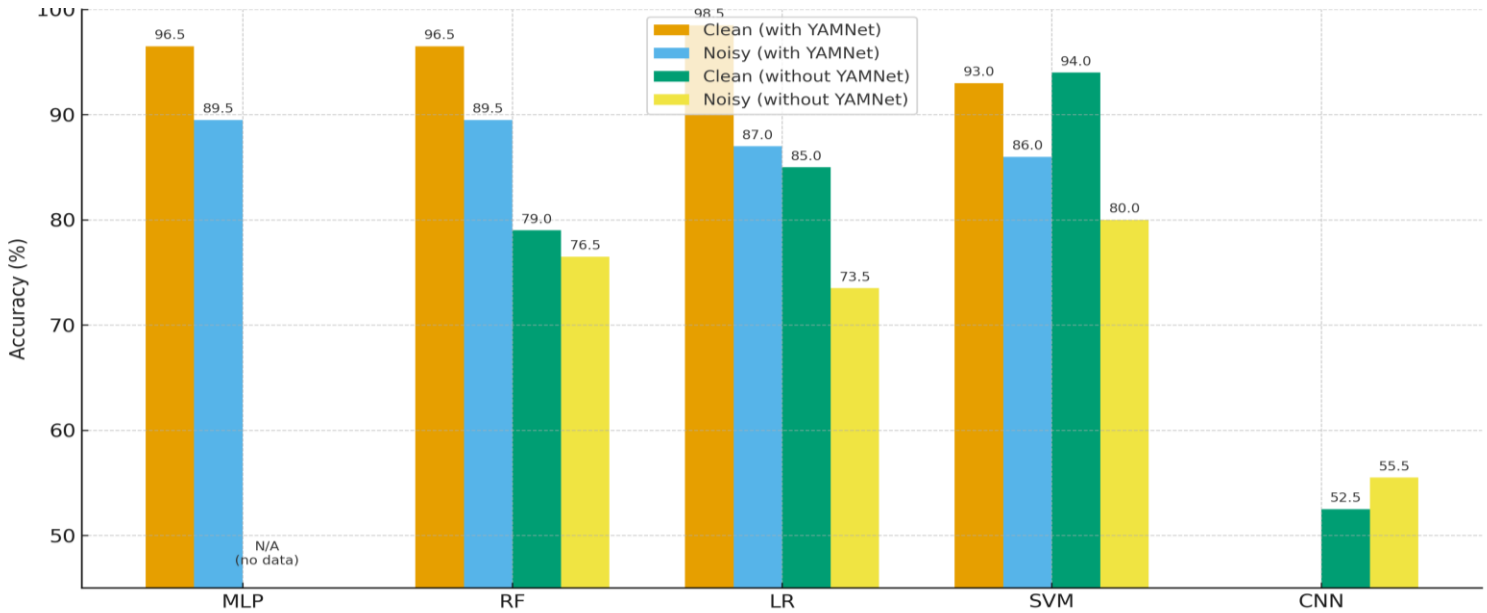
Evaluation Metrics

- Accuracy: % correct predictions
- ROC–AUC: discrimination power
- Confusion matrix: breakdown of errors



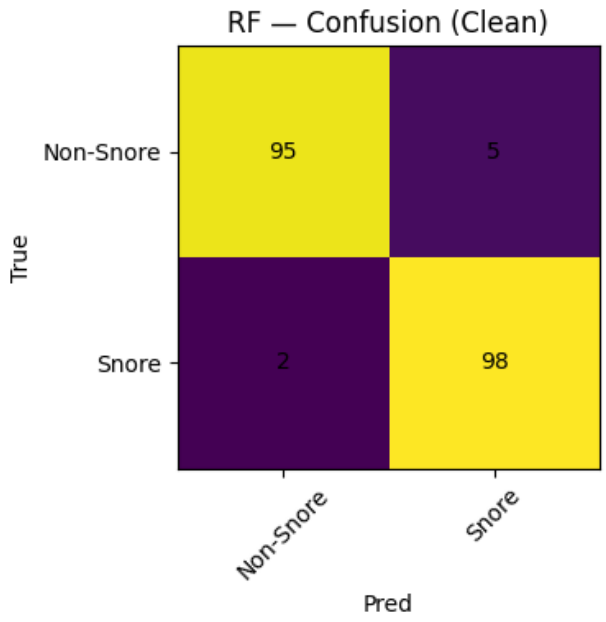
Model Comparisons

- MFCC + LR: 85% clean, 73% noisy
- YAMNet + MLP: 96.5% clean, 89.5% noisy
- Random Forest: robust but heavier
- CNNs: underperformed without large datasets



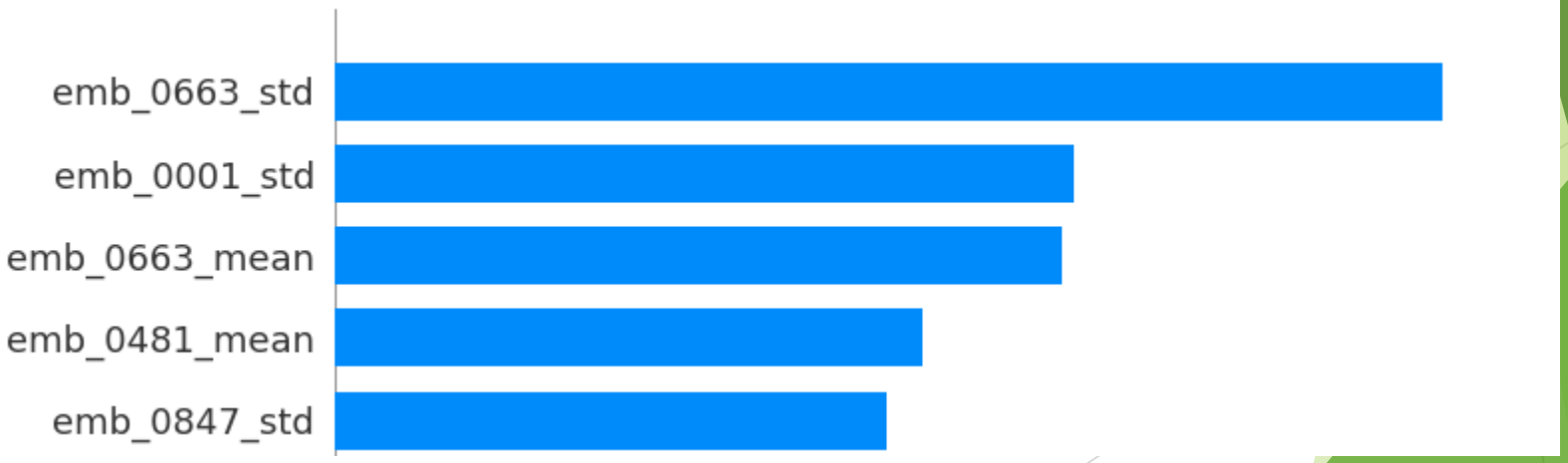
Results – Clean Data

- High accuracy (~93–95%)
- ROC–AUC ~0.95
- Confusion matrix shows few errors



Explainability with SHAP

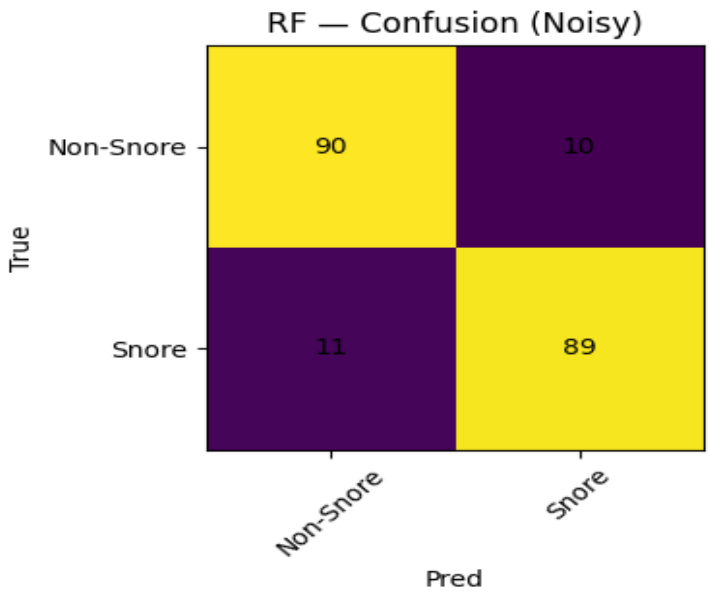
- SHAP showed model uses physiologically meaningful features
- Low-frequency bursts, inhale-exhale cycles
- Invariant to noise due to augmentation



Results – Noisy Data

- Accuracy ~80–85%
- ROC–AUC ~0.85
- More false negatives/positives

Noisy Test



Challenges & Limitations

- Limited data size
- Noise mismatch to real life
- Short isolated clips, not continuous
- Dependence on YAMNet features

Future Work

- Try CNNs on spectrograms
- Collect larger, more varied dataset
- Real-time deployment
- Better noise handling
- Multimodal cues

Ethical & Deployment Considerations

- Dataset bias (age, gender, BMI underrepresentation)
- Privacy: advocate on-device inference
- Edge constraints: low-end devices need pruning/quantization
- Clinical caveats: snore ≠ apnea diagnosis

Conclusion

- Built robust snore classifier with YAMNet + MLP
- >90% accuracy clean, ~80% noisy
- Shows feasibility of AI for sleep health

Q&A

- Questions?